

Computer Vision - Exercise 4

Jun.-Prof. Dr. Kira Maag, Ben Hamscher M.Sc., Arnold Brosch M.Sc.

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This exercise will be discussed the week of 1st of June.

1 Introduction

The following exercises cover Difference of Gaussians, feature matching and perspective projections.

2 Exercises

Exercise 1: Difference of Gaussians (DoG)

The Difference of Gaussians is defined as

$$\text{DoG}(x, y, \sigma) = G(x, y, k\sigma) - G(x, y, \sigma),$$

where G is a Gaussian kernel and $k > 1$.

1. Explain why DoG is a good approximation of the Laplacian of Gaussian.
2. In the DoG pyramid, why are extrema detected in a $3 \times 3 \times 3$ neighborhood?
3. What kind of image structures are detected by DoG extrema?
4. Explain why DoG enables scale-invariant feature detection.

Exercise 2: Feature Matching and Distance Measures

Given two feature descriptors $f_1 \in \mathbb{R}^n$ and $f_2 \in \mathbb{R}^n$, a common similarity measure is the Euclidean distance:

$$d(f_1, f_2) = \|f_1 - f_2\|_2.$$

An improved matching strategy uses the ratio test:

$$r = \frac{\|f_1 - f_2\|}{\|f_1 - f'_2\|},$$

where f'_2 is the second-best match.

1. Why is Euclidean distance a reasonable measure for feature similarity?
2. Explain the intuition behind the ratio test.
3. What does a high ratio value indicate?
4. Why does the ratio test improve robustness against incorrect matches?

Exercise 3: Perspective Projection

A 3D point is given by

$$P = (x, y, z).$$

Using the pinhole camera model, the projected image point is

$$p = (x', y') = \left(-d\frac{x}{z}, -d\frac{y}{z}\right).$$

1. Derive this projection formula using similar triangles.
2. Write the projection in homogeneous coordinates as a matrix multiplication.
3. What information is lost during the projection from 3D to 2D?
4. Explain how depth z influences the projected position.